

A party at Mama Kovacs's home in New York City, December 1972. *Left to right:* Kip Thorne, Margaret Press, Bill Press, Roselyn Teukolsky, and Saul Teukolsky. [Courtesy Sándor J. Kovács.]

Using Teukolsky's equations, one could analyze all sorts of problems: the natural frequencies of black-hole pulsations, the stability of a hole's pulsations, the gravitational radiation produced when a neutron star gets swallowed by a black hole, and more. Such analyses, and extensions of Teukolsky's methods, were immediately undertaken by a small army of researchers: Alexi Starobinsky (a student of Zel'dovich's), Bob Wald (a student of Wheeler's), Jeff Cohen (a student of Dieter Brill's, who was a student of Wheeler's), and many others. Teukolsky himself, with Bill Press, commanded the most important problem: the stability of black-hole pulsations.

Their conclusion, derived by a mixture of computer calculations and calculations with formulas, was disappointing: No matter how fast a black hole spins, its pulsations are stable.[6] The hole's pulsations *do* extract rotational energy from the hole, but they also radiate energy away as gravitational waves; and the rate at which they radiate energy is always greater than the rate they extract it from the hole's spin. Their pulsational energy thus always dies out. It never grows, and the hole therefore cannot be destroyed by its, pulsations.